9. LEPROSY

Overview

Leprosy, also known as **Hansen disease**, is a chronic infectious disease caused mainly by the bacterium **Mycobacterium leprae**. It primarily affects the **skin and peripheral nerves**, but can also affect the mucosa of the upper respiratory tract and the eyes.

Leprosy is **curable**, and early diagnosis and treatment can prevent permanent disability.

Key Facts

- **Causative organism:** *Mycobacterium leprae*
- **Type:** Chronic infectious disease
- **Mainly affects:** Skin and peripheral nerves
- It can also affect the eyes and upper respiratory tract.
- Untreated disease can cause **permanent nerve damage and disability**.
- Leprosy is **curable with multidrug therapy (MDT)**.
- The disease occurs in more than 120 countries.
- India, Brazil and Indonesia continue to report large numbers of new cases.

Transmission

Leprosy is believed to spread mainly through **respiratory droplets from the nose and mouth of an untreated person** following prolonged and close contact.

Important:

Leprosy **does NOT normally spread through casual contact**, such as:

- Shaking hands
- Hugging
- Sharing meals
- Sitting next to someone

Once effective treatment is started, the patient stops transmitting the disease.

Transmission Flow

Untreated infected person
↓
Prolonged, close contact
↓

Respiratory droplets
↓
Transmission to another susceptible person

Symptoms and Signs

Symptoms can develop slowly. The disease commonly affects the skin and peripheral nerves.

Common signs include:

- Pale or reddish **skin patches**
- Loss or reduction of sensation in affected skin
- Numbness
- Thickened or enlarged peripheral nerves
- Muscle weakness
- Loss of sensation in the areas supplied by affected nerves
- Eye involvement
- Nasal/upper respiratory involvement

If left untreated, nerve damage can result in **deformity and permanent disability**.

Diagnosis

WHO identifies **three cardinal signs** that can establish a diagnosis of leprosy:

1. A definite loss of sensation in a pale or reddish skin patch.
2. A thickened/enlarged peripheral nerve associated with loss of sensation and/or muscle weakness.
3. Detection of leprosy bacilli in a **slit-skin smear**.

Classification

For treatment purposes, cases are classified as:

Paucibacillary (PB)

- 1–5 skin lesions
- No bacilli demonstrated in the skin smear.

Multibacillary (MB)

- More than 5 skin lesions, **or**
- Nerve involvement, **or**
- Bacilli demonstrated in a slit-skin smear.

Treatment

Leprosy is **curable**.

WHO recommends **multidrug therapy (MDT)** consisting of:

- **Dapsone**
- **Rifampicin**
- **Clofazimine**

Recommended duration:

- **PB:** 6 months
- **MB:** 12 months

Early diagnosis and prompt treatment help prevent disability.

Important

Treatment should be taken under the guidance of a healthcare professional. Completing the prescribed treatment is essential.

Prevention and Control

Prevention focuses on **early detection, treatment and contact screening**.

WHO recommends:

- Early detection of cases
- Prompt initiation of MDT
- Screening of household, neighbourhood and social contacts
- Single-dose rifampicin as **post-exposure prophylaxis (SDR-PEP)** for eligible contacts according to WHO recommendations.

Complications

Untreated leprosy can result in:

- Permanent nerve damage
- Muscle weakness
- Loss of sensation
- Deformities
- Disability
- Eye complications
- Social stigma and discrimination.

When to Seek Medical Care

Medical evaluation should be sought when a person develops:

- A persistent pale or reddish skin patch with reduced sensation
- Numbness
- Unexplained weakness of the hands or feet
- Thickened nerves
- Persistent skin changes associated with sensory loss
- Eye symptoms associated with suspected leprosy

Early diagnosis and treatment are important because nerve damage can become permanent.

Quick Fact Sheet

Feature	**Leprosy**
Other name	Hansen disease
Cause	*Mycobacterium leprae*
Main organs affected	Skin and peripheral nerves
Spread	Prolonged, close contact with untreated cases; respiratory droplets

Feature	Leprosy
Main signs	Skin patches with sensory loss, nerve thickening, weakness
Diagnosis	Clinical signs ± slit-skin smear
Treatment	Multidrug therapy
Main medicines	Dapsone + Rifampicin + Clofazimine
PB treatment	6 months
MB treatment	12 months
Prevention	Early detection, treatment and contact screening
Outcome	Curable; early treatment helps prevent disability

Official Reference

World Health Organization (WHO). “Leprosy.” Updated 23 January 2026.

10. CHOLERA

Overview

Cholera is an **acute diarrhoeal infection** caused by consuming food or water contaminated with the bacterium **Vibrio cholerae**.

It can cause severe acute watery diarrhoea and **rapid dehydration**. Without treatment, severe cholera can become fatal within hours. However, most cases can be successfully treated with **oral rehydration solution (ORS)**.

Key Facts

- **Causative organism:** *Vibrio cholerae*
- **Main route of transmission:** Contaminated food or water
- **Main symptom:** Acute watery diarrhoea
- Severe disease can cause rapid dehydration.
- Most infections produce no or mild symptoms.
- Severe cases require rapid fluid replacement.
- Safe water, sanitation and hygiene are central to prevention.
- Oral cholera vaccines can help prevent and control outbreaks.

Transmission

Cholera is mainly transmitted through the **faecal–oral route**.

Typical transmission cycle:

Infected person → Faeces contaminate environment → Water/food contamination → Another person consumes contaminated water/food → Infection

Poor sanitation and unsafe drinking water increase the risk of transmission.

Symptoms

Symptoms generally appear **12 hours to 5 days after infection**.

Common symptoms:

- **Acute watery diarrhoea**
- Vomiting

- Rapid loss of body fluids
- Dehydration
- Weakness

Most infected people develop no symptoms or only mild/moderate diarrhoea.

A minority develop **severe acute watery diarrhoea with life-threatening dehydration**.

Dehydration — Major Danger

The most serious complication of cholera is **severe dehydration** caused by rapid fluid loss.

Signs of significant dehydration can include:

- Extreme thirst
- Weakness
- Reduced urination
- Dry mouth
- Sunken eyes
- Dizziness
- Rapid deterioration

Severe dehydration is a medical emergency.

Diagnosis

Cholera should be suspected in a person with **acute watery diarrhoea**, particularly during an outbreak or in an area where cholera transmission is occurring.

Laboratory testing can help confirm the presence of *Vibrio cholerae*.

WHO emphasizes the importance of **epidemiological and laboratory surveillance** for detecting and monitoring outbreaks.

Treatment

The most important principle:

Replace lost fluids immediately.

Mild/Moderate illness

- **Oral rehydration solution (ORS)** is the main treatment.
- Continue appropriate feeding/nutrition.
- Monitor for worsening dehydration.

Severe illness

Severe cases require:

- **Intravenous fluids**
- ORS
- **Antibiotics**, when indicated

Rapid treatment is critical because severe cholera can progress very quickly.

Simple Treatment Flow

Watery diarrhoea
↓
Assess dehydration
↓
No/severe dehydration → ORS and monitoring
Severe dehydration → IV fluids + ORS + appropriate antibiotics

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Prevention

Prevention focuses strongly on **WASH — Water, Sanitation and Hygiene**.

Safe water

- Drink safe water.
- Use appropriately treated water.
- Avoid potentially contaminated drinking water.

Food safety

- Eat properly prepared food.
- Keep food protected from contamination.
- Maintain hand hygiene before preparing or eating food.

Sanitation

- Use safe sanitation facilities.
- Dispose of human waste safely.
- Prevent contamination of drinking-water sources.

Vaccination

Oral cholera vaccines (OCV) can help prevent and control cholera, particularly as part of outbreak-control and high-risk-area strategies.

Complications

The major complication is:

Severe dehydration → Shock → Death

Other consequences may include:

- Electrolyte disturbances
- Kidney injury associated with severe dehydration
- Circulatory collapse

Rapid rehydration can prevent most deaths.

When to Seek Urgent Care

Seek medical attention immediately for:

- Profuse watery diarrhoea
- Repeated vomiting
- Signs of dehydration
- Severe weakness
- Reduced urine
- Dizziness or fainting
- Rapid worsening of symptoms

Quick Fact Sheet

Feature	Cholera
Cause	*Vibrio cholerae*
Spread	Contaminated food/water
Incubation	12 hours–5 days
Main symptom	Acute watery diarrhoea
Major danger	Severe dehydration
First-line treatment	ORS
Severe cases	IV fluids + ORS + antibiotics
Prevention	Safe water, sanitation, hygiene
Vaccine	Oral cholera vaccine
Outcome	Highly treatable when treated promptly

Official Reference

World Health Organization (WHO). “Cholera.” 5 December 2024

11. JAPANESE ENCEPHALITIS

Overview

Japanese encephalitis (JE) is a **mosquito-borne viral disease** that causes inflammation of the brain.

It is caused by **Japanese encephalitis virus (JEV)**, a flavivirus related to dengue, yellow fever and West Nile viruses.

Most JEV infections are asymptomatic or cause only mild illness. However, a small proportion progress to severe encephalitis, which can cause permanent neurological disability or death.

Key Facts

- **Causative agent:** Japanese encephalitis virus (JEV)
- **Virus family:** Flavivirus
- **Main vector:** *Culex* mosquitoes
- **Mainly affects:** Central nervous system/brain
- Most infections are mild or asymptomatic.
- Severe disease can cause encephalitis, seizures, paralysis and coma.
- The majority of clinical cases occur in **children below 15 years**.
- There is **no specific antiviral cure**.
- Effective vaccines are available for prevention.

Transmission

JEV is mainly transmitted by infected **Culex mosquitoes**, particularly *Culex tritaeniorhynchus*.

The virus circulates naturally between:

Mosquitoes ↔ pigs / water birds

Humans become infected when bitten by an infected mosquito.

Humans generally do not develop enough virus in their blood to infect feeding mosquitoes, so humans are considered **dead-end hosts**.

Transmission Flow

JEV-infected mosquito
↓
Mosquito bites human
↓
JEV infection
↓
Usually asymptomatic/mild infection
OR
Severe encephalitis

Symptoms

The incubation period is approximately **4–14 days**.

Most infections produce:

- No symptoms, or
- Mild fever
- Headache

In children, **gastrointestinal pain and vomiting** may be important early symptoms.

Severe Japanese Encephalitis

Severe disease may develop rapidly.

Warning symptoms include:

- High fever
- Severe headache
- Neck stiffness
- Disorientation
- Altered consciousness
- Seizures
- Spastic paralysis
- Coma

Severe encephalitis is a **medical emergency**.

Complications

Among people who develop encephalitis:

- Case-fatality can be **as high as 30%**.
- Around **30–50% of survivors** may develop permanent neurological, cognitive or behavioural problems.

Possible long-term problems include:

- Seizures
- Hearing loss
- Vision problems
- Speech/language difficulties
- Memory problems
- Communication difficulties
- Weakness or paralysis
- Cognitive and behavioural changes.

Diagnosis

A person living in or travelling to a JE-endemic area who develops **encephalitis** may be suspected of having JE.

Diagnosis involves:

- Clinical assessment
- Evaluation of neurological symptoms
- **Lumbar puncture**, where appropriate
- Laboratory confirmation

WHO recommends detection of **JEV-specific IgM antibodies** using an IgM-capture ELISA.

Testing of **cerebrospinal fluid (CSF)** is preferred because it can reduce false-positive results associated with previous infection or vaccination.

Treatment

There is currently **no specific antiviral treatment** for Japanese encephalitis.

Treatment is supportive and focuses on:

- Stabilizing the patient
- Managing severe symptoms
- Supporting breathing and circulation when necessary
- Managing seizures and other complications
- Monitoring neurological status
- Providing appropriate intensive care when required.

Important

Suspected encephalitis requires urgent medical attention.

Prevention

1. Vaccination

Vaccination is the **most important preventive measure** in populations where JE is a public-health concern.

WHO recommends JE vaccination as part of national immunization programmes in areas where JE is recognized as a public-health priority.

2. Mosquito-bite prevention

- Use mosquito repellents.
- Wear clothing that reduces exposed skin.
- Use mosquito nets where appropriate.
- Reduce mosquito breeding sites.
- Take additional precautions in areas with active JE transmission.

High-Risk Settings

JE transmission is particularly associated with:

- Rural and periurban environments
- Areas with rice cultivation

- Areas where pigs and water birds are present
- Warm/tropical areas where mosquito populations increase

Transmission may intensify during **rainy seasons** in tropical and subtropical regions.

When to Seek Urgent Medical Attention

Immediate medical care is required when fever is accompanied by:

- Severe headache
- Neck stiffness
- Confusion/disorientation
- Altered consciousness
- Seizures
- Weakness/paralysis
- Coma
- Rapid neurological deterioration

Quick Fact Sheet

Feature	**Japanese Encephalitis**
Cause	Japanese encephalitis virus
Virus type	Flavivirus
Vector	Mainly *Culex* mosquitoes
Incubation	4–14 days
Main organ affected	Brain/CNS
Common mild symptoms	Fever, headache
Severe symptoms	Encephalitis, seizures, paralysis, coma
Specific antiviral treatment	None
Treatment	Supportive care

Feature	Japanese Encephalitis
Major prevention	Vaccination + mosquito protection
Severe disease fatality	Up to 30%
Long-term sequelae among survivors	Approximately 30–50%

Official Reference

World Health Organization (WHO). “Japanese encephalitis.” 6 August 2024.

12. KALA-AZAR (VISCERAL LEISHMANIASIS)

Overview

Kala-azar, also known as **visceral leishmaniasis (VL)**, is the most serious form of leishmaniasis.

It is caused by **Leishmania parasites** and transmitted through the bite of infected **female phlebotomine sandflies**.

If untreated, visceral leishmaniasis can be fatal.

Key Facts

- **Other name:** Visceral leishmaniasis / Kala-azar
- **Cause:** *Leishmania* parasites
- **Vector:** Infected female phlebotomine sandflies
- **Main organs affected:** Spleen, liver and blood-forming system
- **Major symptoms:** Prolonged/irregular fever, weight loss, enlarged spleen and liver, anaemia
- VL is the **most severe form of leishmaniasis**.
- Untreated VL is potentially fatal.
- It is **treatable and curable** when properly diagnosed and treated.

Transmission

The parasite is transmitted through the bite of an infected **female sandfly**.

Transmission Flow

Infected sandfly
↓
Bites human
↓
Leishmania parasites enter the body
↓
Parasites multiply in tissues
↓
Visceral leishmaniasis

The disease is associated with factors such as:

- Malnutrition
- Population displacement
- Poor housing
- Weak immune systems
- Poverty and limited access to healthcare.

Symptoms

The main clinical features of visceral leishmaniasis include:

- **Irregular bouts of fever**
- Significant **weight loss**
- **Enlarged spleen (splenomegaly)**
- **Enlarged liver (hepatomegaly)**
- **Anaemia**
- Weakness and fatigue

Symptoms can develop gradually and may resemble other diseases, so appropriate medical evaluation is important.

Why Kala-Azar Is Serious

The disease affects internal organs and the body's immune and blood-forming systems.

Without treatment, visceral leishmaniasis can become **life-threatening**.

WHO states that VL is fatal in **over 95% of cases when untreated**.

Diagnosis

Diagnosis of visceral leishmaniasis involves combining:

Clinical assessment

Look for:

- Prolonged/irregular fever
- Weight loss

- Enlarged spleen
- Enlarged liver
- Anaemia

Laboratory confirmation

WHO recommends combining clinical findings with:

- **Parasitological tests**
- **Serological tests**, including rapid diagnostic tests where appropriate.

Simple Diagnostic Flow

Prolonged fever + weight loss + enlarged spleen/liver
↓
Clinical suspicion of VL
↓
Parasitological / serological testing
↓
Diagnosis confirmed
↓
Prompt complete treatment

Treatment

Visceral leishmaniasis is **treatable and curable**, but patients require appropriate medical treatment.

Treatment depends on factors including:

- Geographic location
- Leishmania species
- Clinical condition
- Coexisting diseases
- Immune status.

Important

All patients diagnosed with visceral leishmaniasis require prompt and complete treatment.

Treatment should be selected and administered by qualified healthcare professionals.

Prevention

Because transmission occurs through sandflies, prevention focuses on reducing exposure to infected sandflies.

Personal protection

- Use insect repellents.
- Wear clothing that minimizes exposed skin.
- Use protective nets where appropriate.
- Reduce exposure to sandflies, particularly in endemic areas.

Environmental control

- Improve housing conditions.
- Reduce sandfly breeding/resting environments where feasible.
- Support community vector-control programmes.

Public-health measures

- Early detection
- Prompt treatment
- Surveillance
- Vector control
- Community awareness

Risk Factors

Risk is associated with:

- Poverty
- Malnutrition
- Poor housing
- Population displacement
- Weak immune system
- Living in or travelling to endemic areas.

Complications

Untreated or inadequately treated VL can lead to:

- Severe anaemia
- Progressive weakness and wasting
- Severe systemic illness
- Death

A further complication can occur after treatment in some endemic settings: **post-kala-azar dermal leishmaniasis (PKDL)**, which is characterized by skin manifestations following VL. This requires appropriate clinical evaluation and management.

When to Seek Medical Care

Medical evaluation should be sought when a person in or from an endemic area develops:

- Persistent or prolonged fever
- Unexplained weight loss
- Enlarged abdomen due to an enlarged spleen/liver
- Severe weakness
- Pallor or suspected anaemia
- Other symptoms suggestive of visceral leishmaniasis

Early diagnosis and complete treatment are essential.

Quick Fact Sheet

Feature	**Kala-Azar / Visceral Leishmaniasis**
Cause	*Leishmania* parasites
Vector	Female phlebotomine sandfly
Main form	Visceral leishmaniasis
Main symptoms	Fever, weight loss, enlarged spleen/liver, anaemia

Feature	Kala-Azar / Visceral Leishmaniasis
Major organs affected	Spleen, liver and blood-forming system
Diagnosis	Clinical + parasitological/serological testing
Treatment	Specific anti-leishmanial treatment
Prognosis	Curable with appropriate treatment
Untreated disease	Potentially fatal
Prevention	Sandfly-bite prevention + vector control
Major risk factors	Malnutrition, poverty, poor housing, displacement, immunosuppression

Official Reference

World Health Organization (WHO). “Leishmaniasis.”